Practice Questions

1. Let Data Encryption Standard (DES) be a block cipher where encryption of plaintext x with key K is denoted by

$$y=DES(x,K)$$

Let $c(\cdot)$ represent the **bitwise complement** operation. Define:

$$y'=DES(c(x),c(K))$$

Prove that:

$$y'= c(y)$$

That is, show that **complementing both the plaintext and the key results in the complement of the ciphertext**.

Note: Your proof should rely only on the **high-level structure of DES** (such as Feistel structure and XOR operations). The internal details of S-boxes or permutations are not required.

2. Consider the Data Encryption Standard (DES) algorithm. Suppose the round function F(R,K) is modified such that it ignores the key K and behaves in one of the following ways:

(a) F(R,K) = 1 (a 32-bit string of all zeros)
(b) F(R,K)=R'

For each case, answer the following:

What overall function does DES compute after 16 rounds of encryption?

3. There are four weak DES keys what are they? What is the likelihood that randomly selected key is weak?

4. Consider performing a known-plaintext attack against the Data Encryption Standard (DES) using a single pair of plaintext and cipher text.

(a) In the worst-case scenario, how many keys are required to be tested if an exhaustive key search is employed in a straightforward manner?

(b) In the average-case scenario, how many keys need to be tested if exhaustive search is employed in a straightforward manner?

5. Suppose that we have the following 128-bit AES key, given in hexadecimal notation:

2B7E151628AED2A6ABF7158809CF4F3C

Construct the complete key schedule arising from this key.

6. Compute the encryption of the following plaintext (given in hexadecimal notation) using the 10-round AES:

3243F6A8885A308D313198A2E0370734

Use the 128-bit key from the previous exercise.

7. Show that $S1(x1) \oplus S1(x2) \neq S1(x1 \oplus x2)$, where "$\oplus$" denotes bitwise XOR and S1(x1)= output of the S-box 1 for x1 for:

(1) x1 = 000000, x2 = 000001

(2) x1 = 111111, x2 = 100000

(3) x1 = 101010, x2 = 010101

S-box S_1

S_1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	01	10	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13